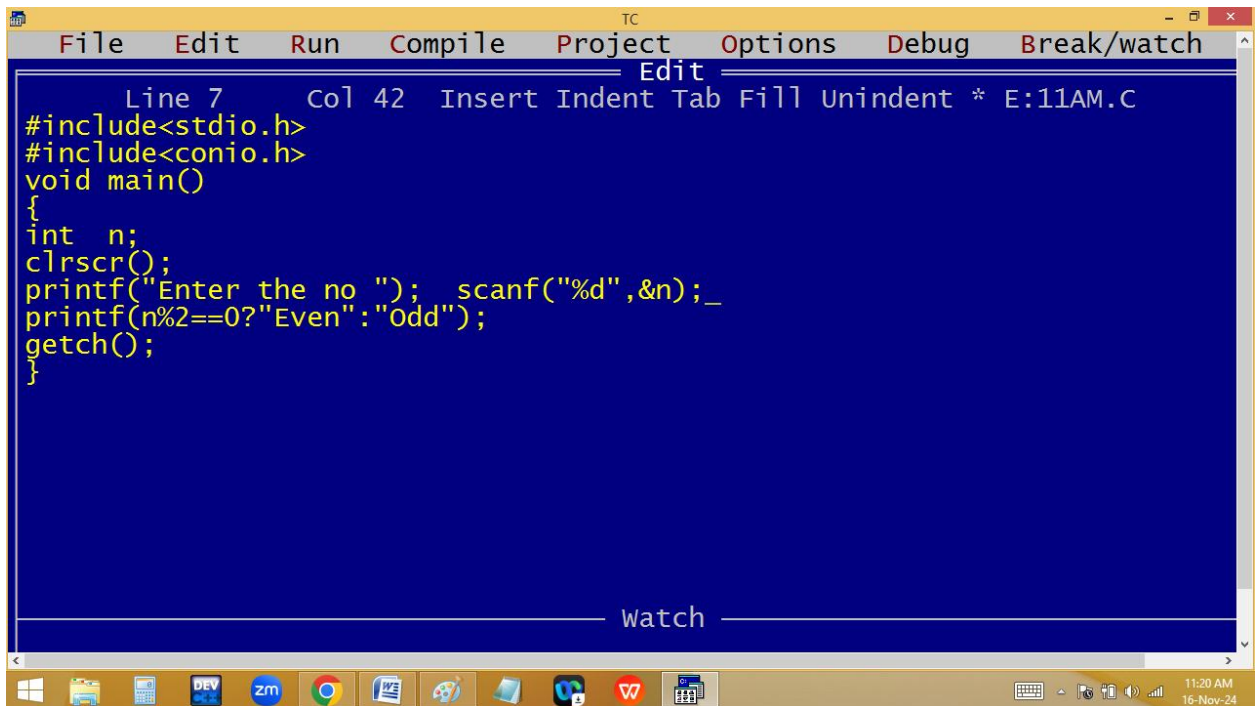


## Ternary operator / conditional operator [ ? : ]

It is using 3 operands.

conditional part ? True part : false part ;

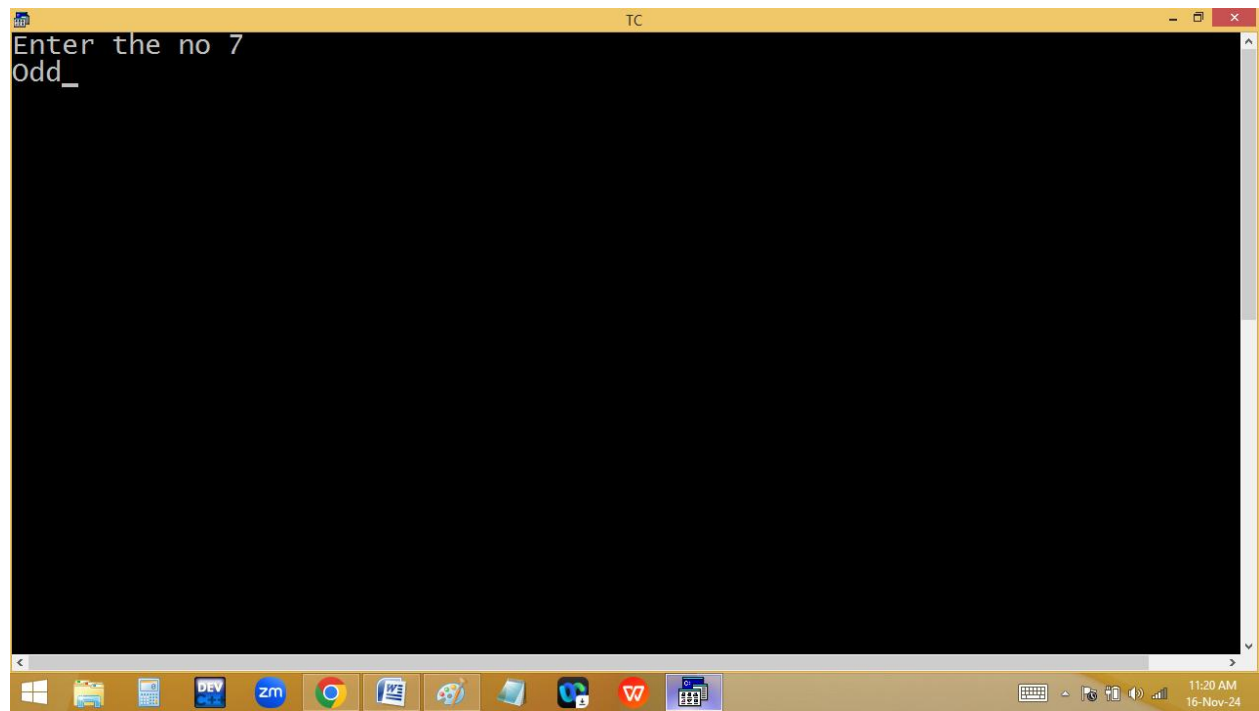
### Finding even/odd using ternary operator



```
TC
File Edit Run Compile Project Options Debug Break/watch
Edit
Line 7 Col 42 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
    int n;
    clrscr();
    printf("Enter the no "); scanf("%d",&n);_
    printf(n%2==0?"Even":"Odd");
    getch();
}
```

Watch

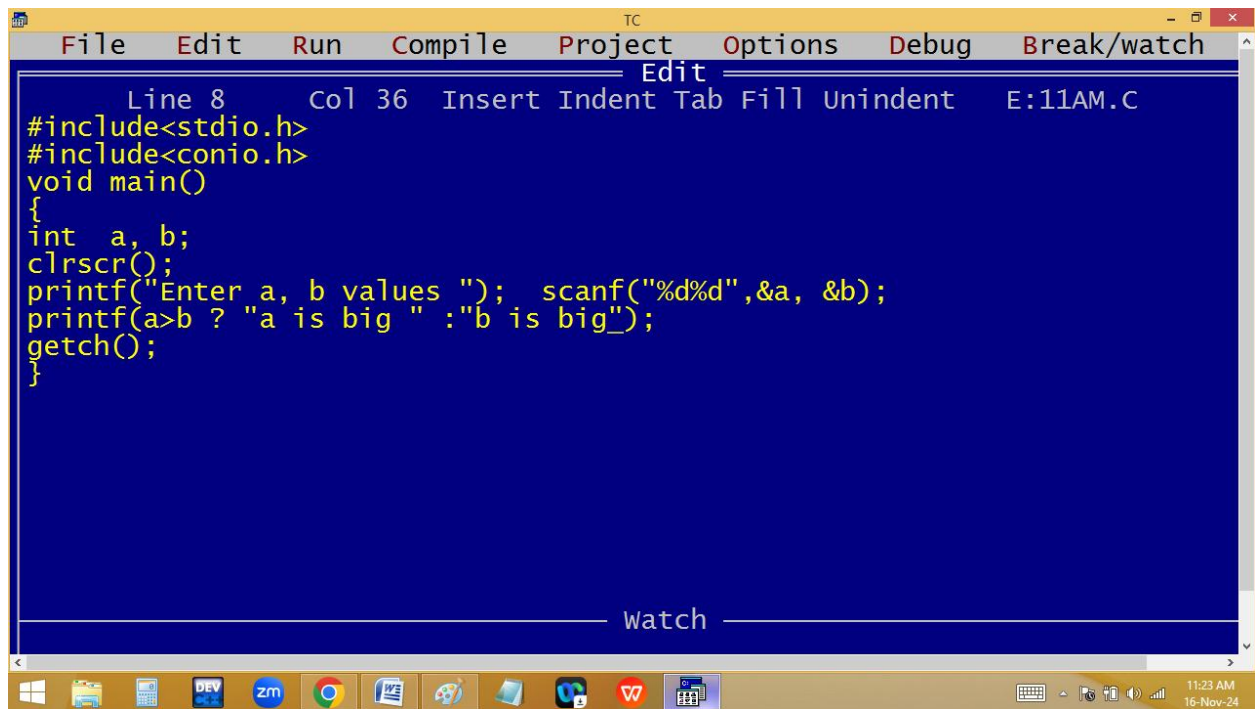
11:20 AM 16-Nov-24



```
TC
Enter the no 8
Even_
```

$8\%2=0==0$  →  
`printf( n%2==0 ? " Even " : " Odd " );`  
 $9\%2=1==0$  →

Finding max in 2 no's using ternary operator:



The screenshot shows the Turbo C++ IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a toolbar. The main window displays a C program with the following code:

```
Line 8 Col 36 Insert Indent Tab Fill Unindent E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a, b;
clrscr();
printf("Enter a, b values "); scanf("%d%d",&a, &b);
printf(a>b ? "a is big " : "b is big");
getch();
}
```

The status bar at the bottom indicates the file name is E:11AM.C and the current time is 11:23 AM on 16-Nov-24.

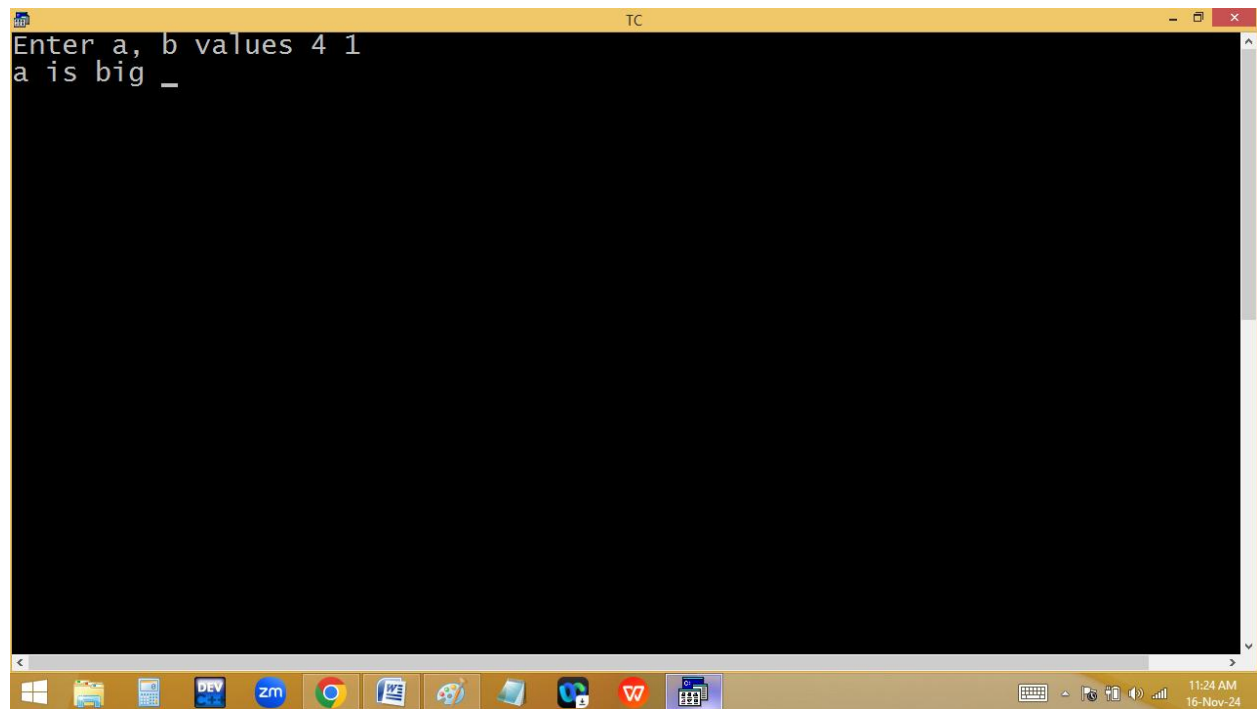


The screenshot shows the Turbo C++ IDE with the same menu bar and toolbar. The main window displays the output of the program:

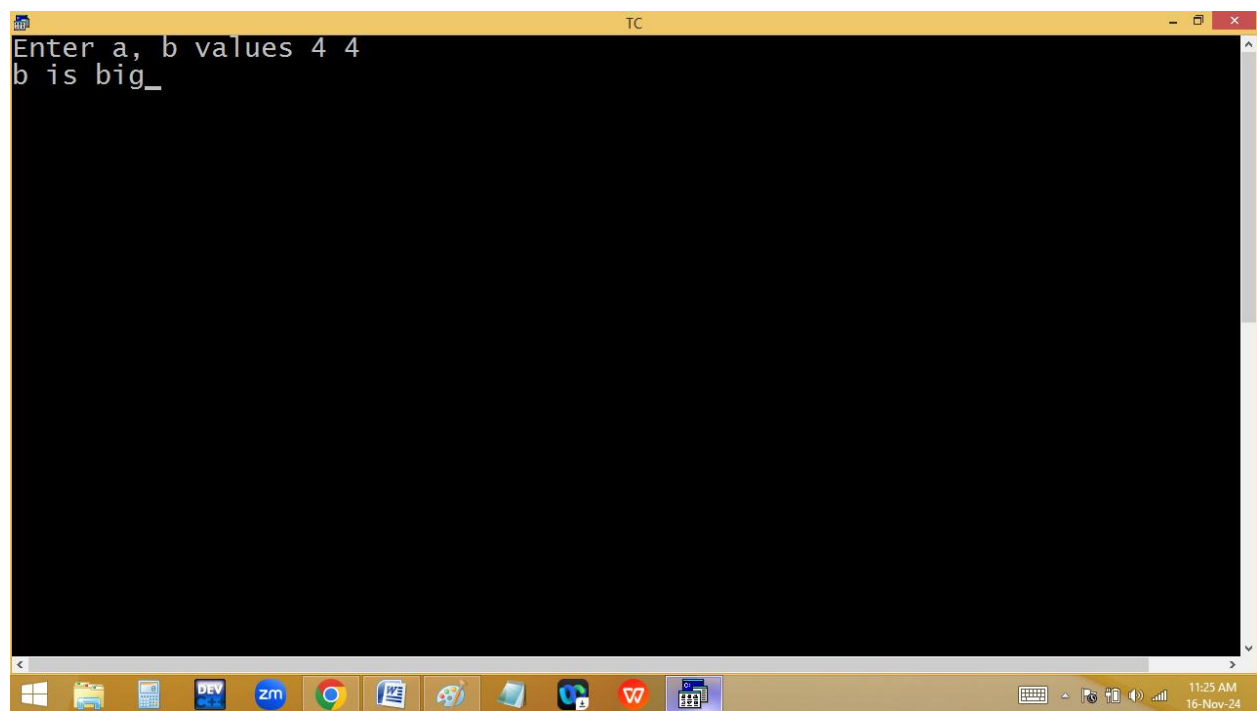
```
Enter a, b values 1 2
b is big
```

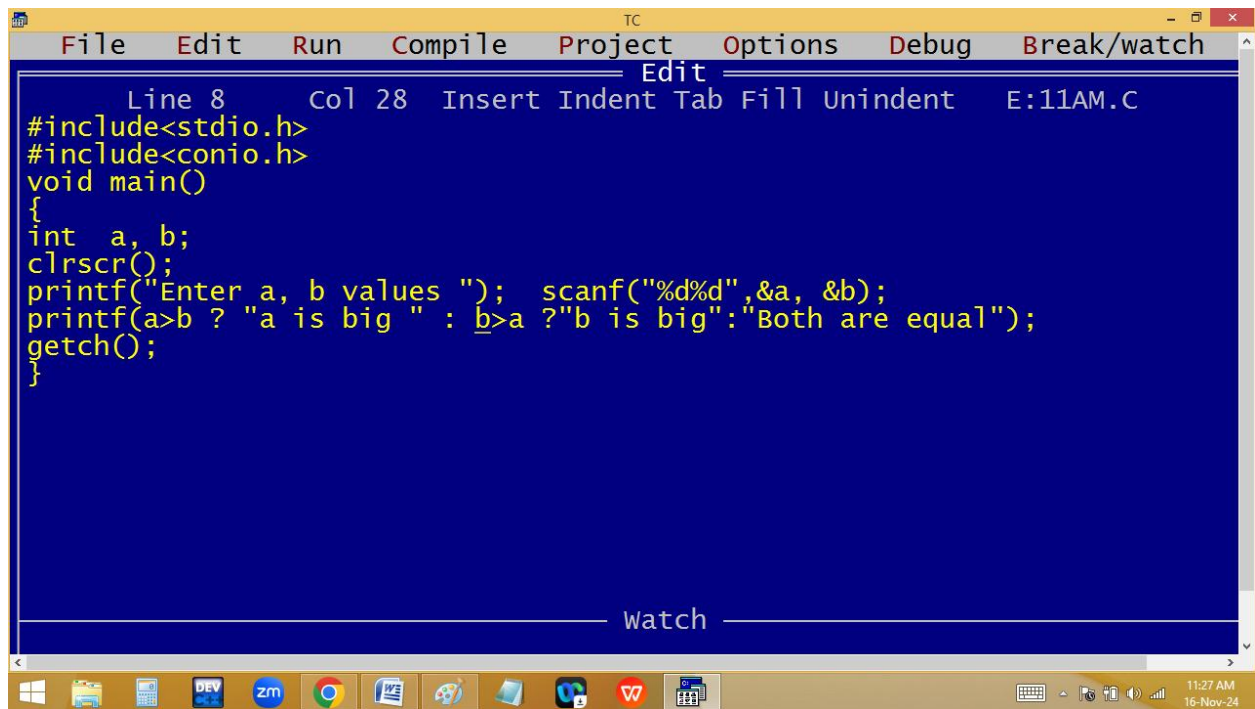
The status bar at the bottom indicates the current time is 11:24 AM on 16-Nov-24.

```
TC
Enter a, b values 4 1
a is big _
```

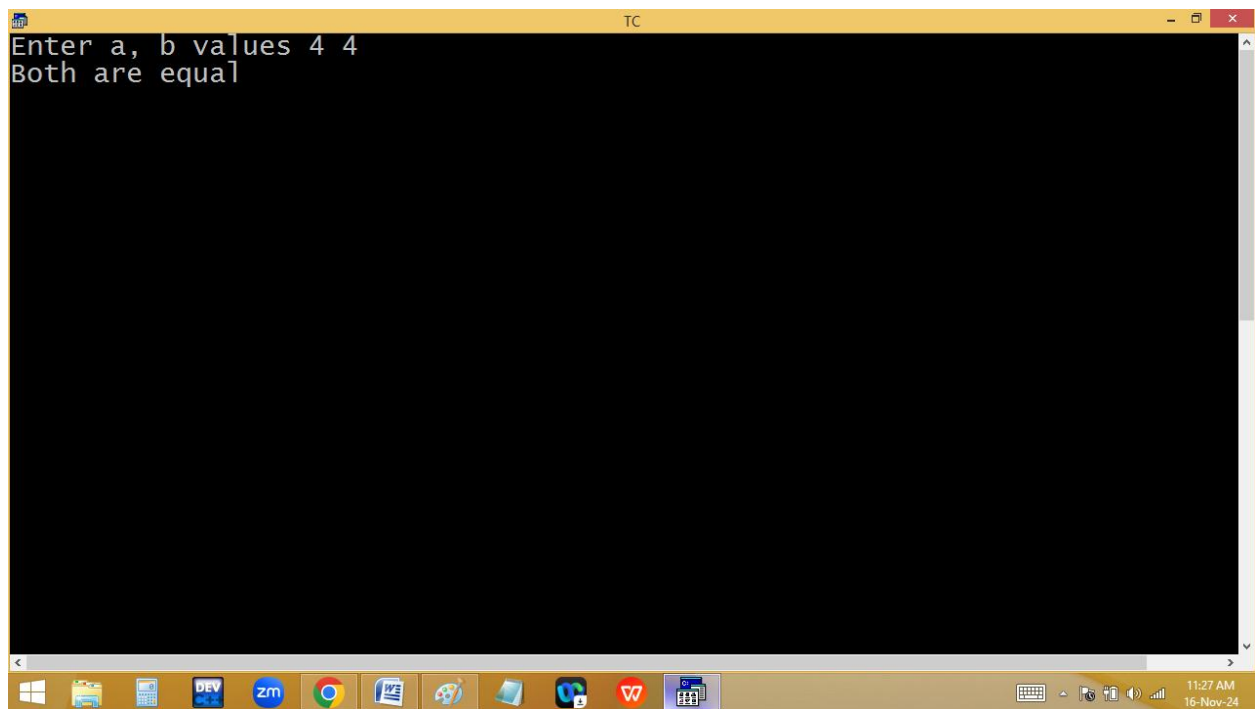


```
TC
Enter a, b values 4 4
b is big _
```

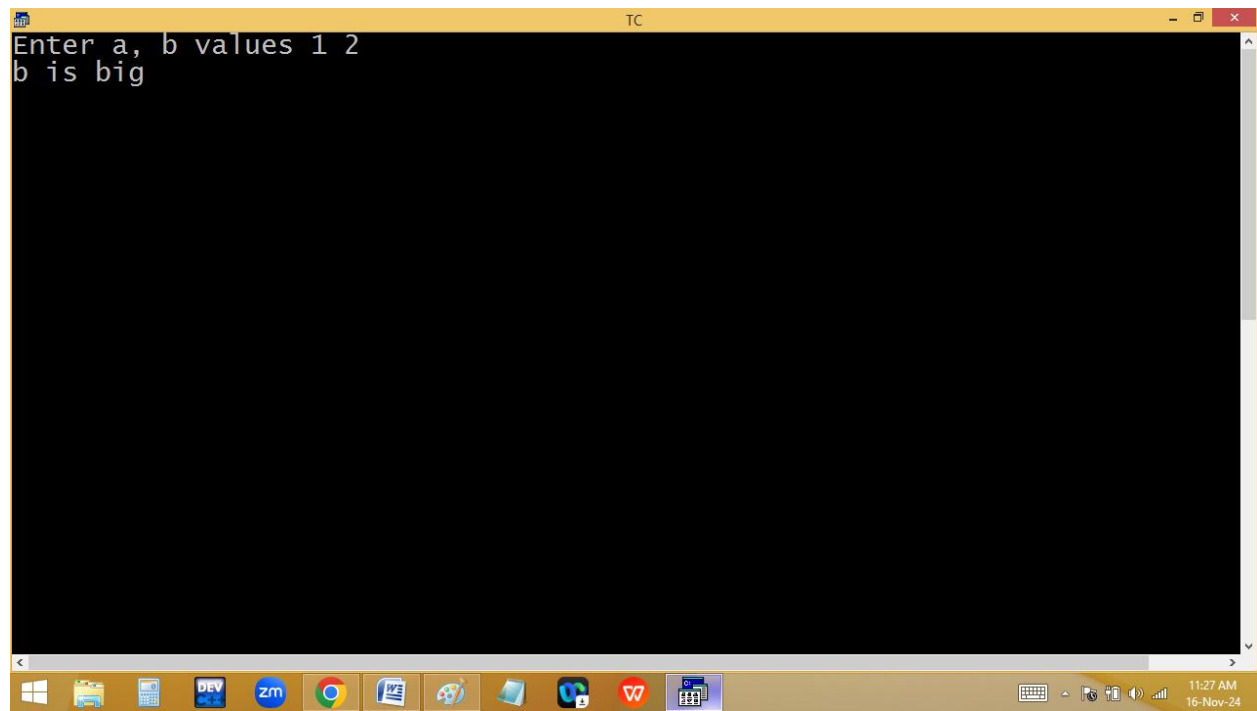




```
Line 8 Col 28 Insert Indent Tab Fill Unindent E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a, b;
clrscr();
printf("Enter a, b values "); scanf("%d%d",&a, &b);
printf(a>b ? "a is big " : b>a ? "b is big ":"Both are equal");
getch();
}
```



```
Enter a, b values 4 4
Both are equal
```

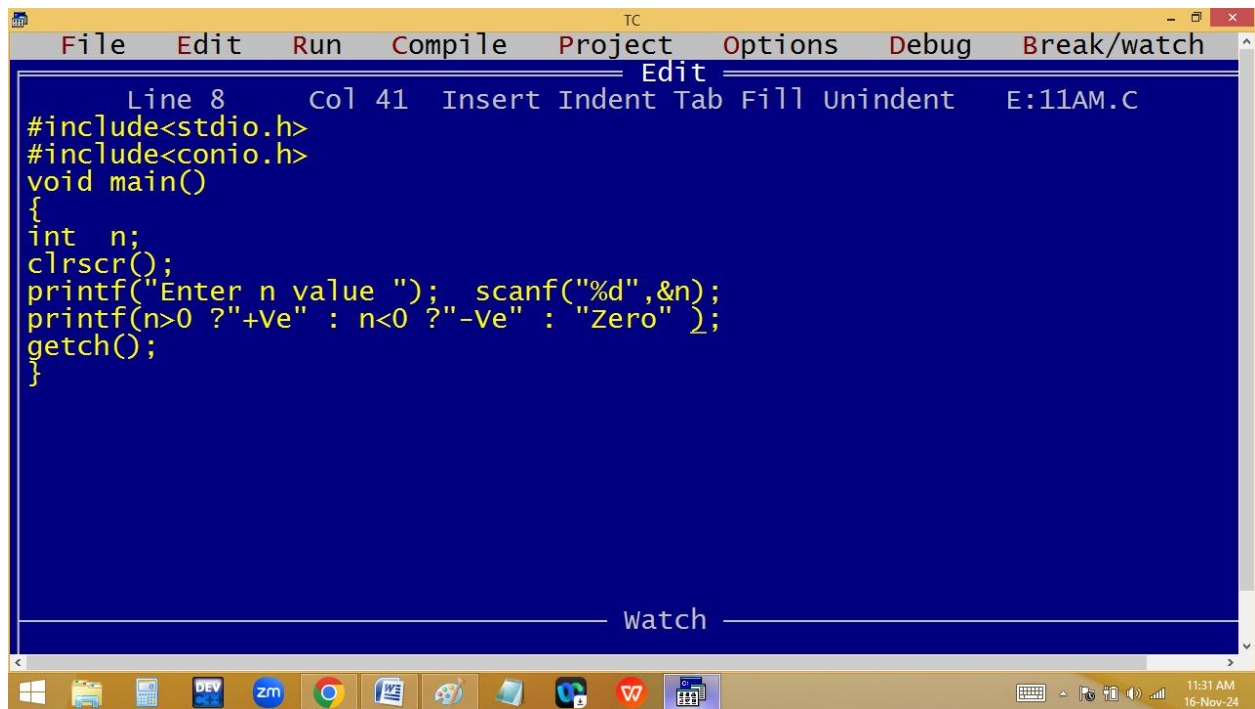


```
TC
Enter a, b values 2 1
a is big _
```

$4 > 4$    
`printf( a>b ? " a is big " : b>a ? "b is big" : "Both are equal " );`  
 $4 > 4$  

Finding +Ve / -Ve / 0:





TC

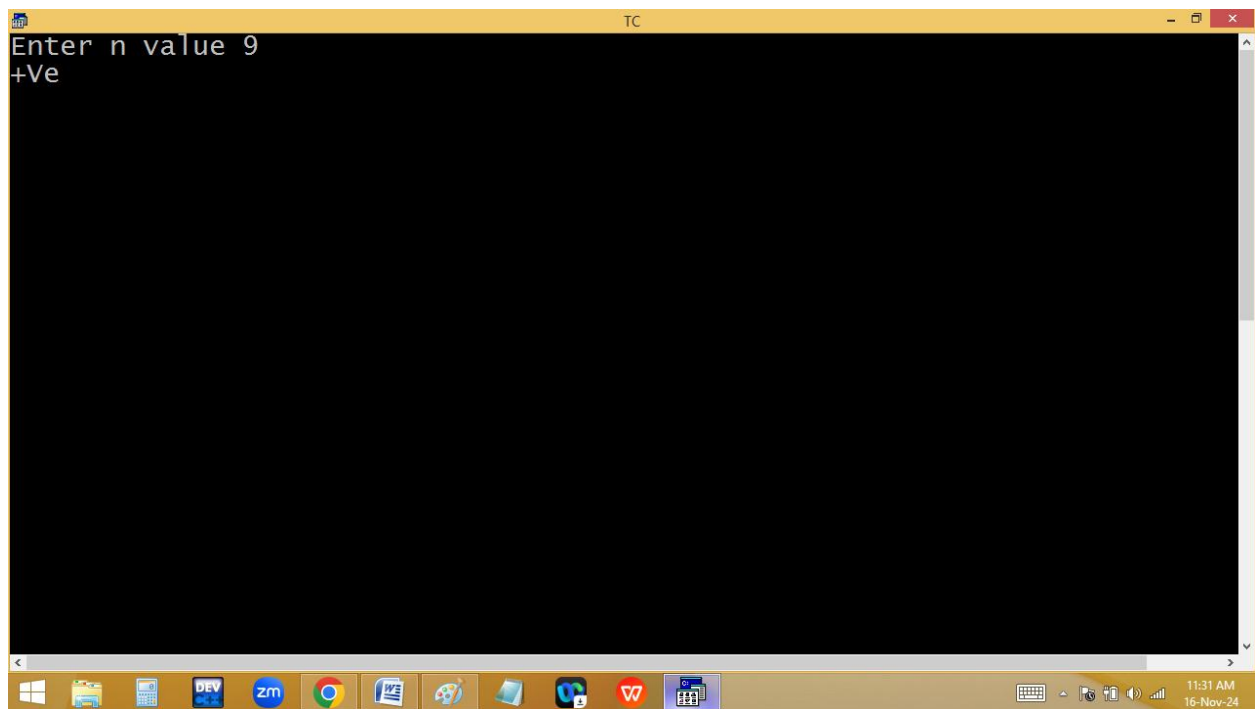
File Edit Run Compile Project Options Debug Break/watch

Line 8 Col 41 Insert Indent Tab Fill Unindent E:11AM.C

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int n;
    clrscr();
    printf("Enter n value "); scanf("%d",&n);
    printf(n>0 ?"+ve" : n<0 ?"-ve" : "Zero" );
    getch();
}
```

Watch

11:31 AM 16-Nov-24



TC

```
Enter n value 9
+ve
```

11:31 AM 16-Nov-24

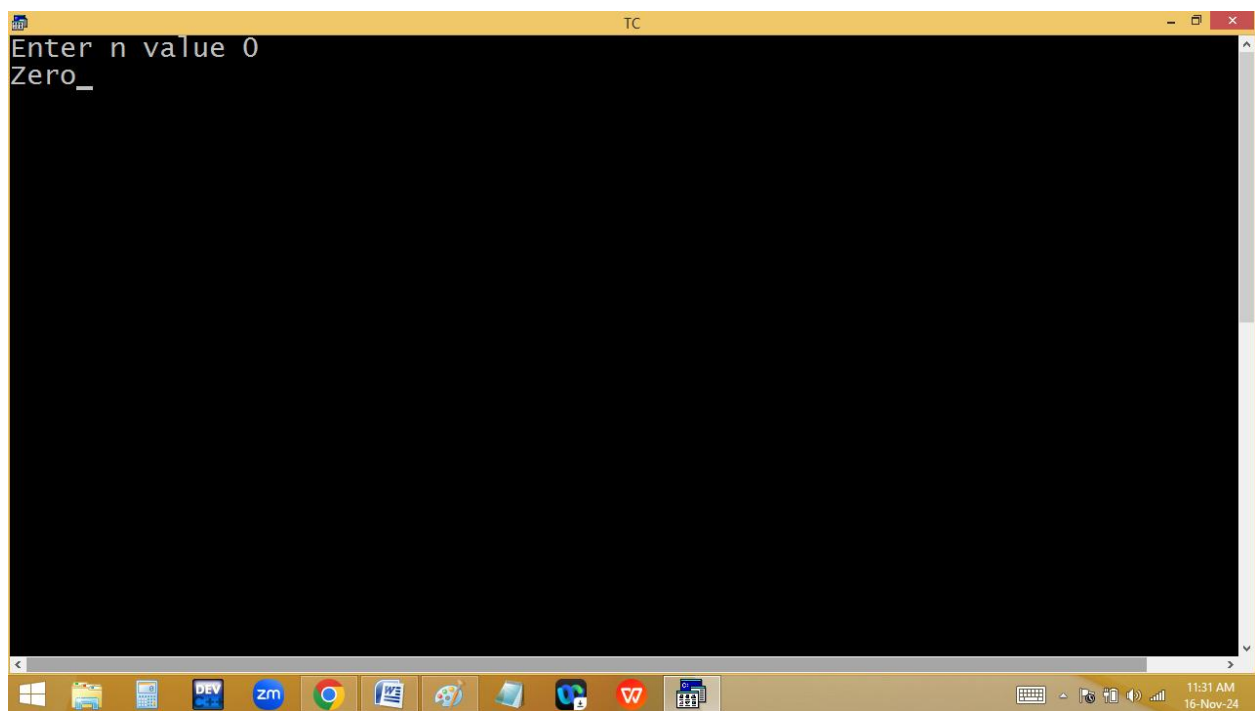
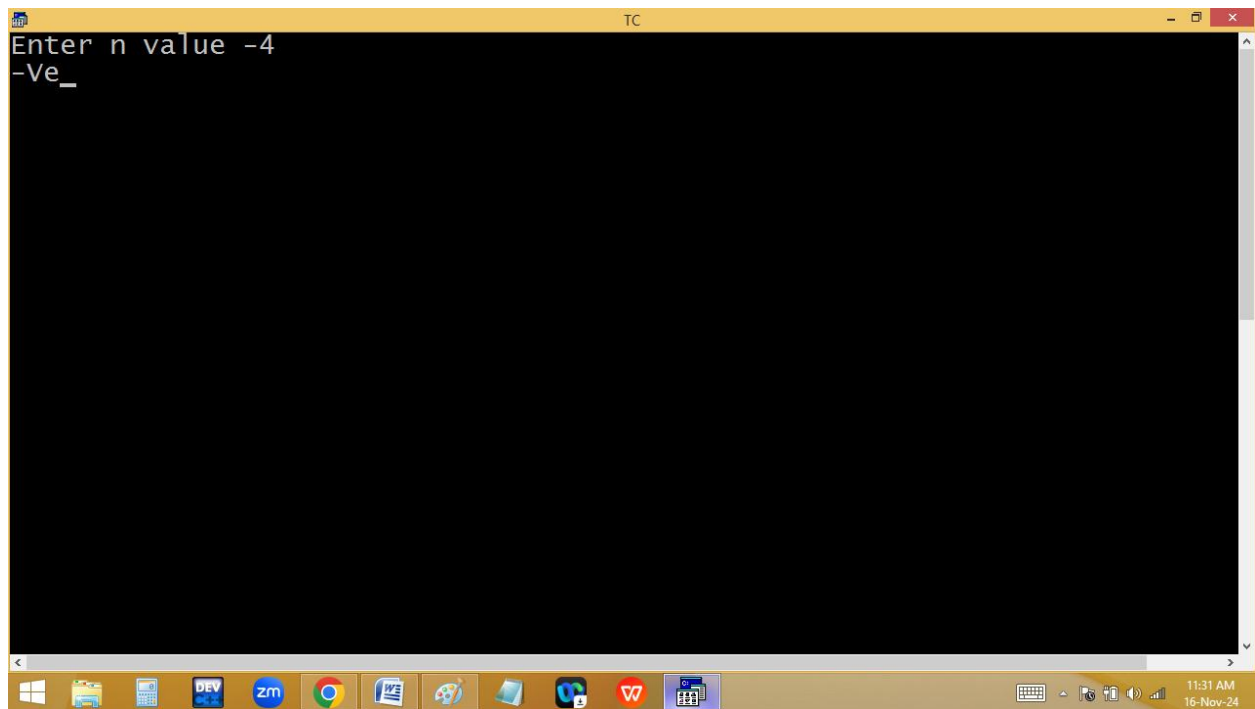


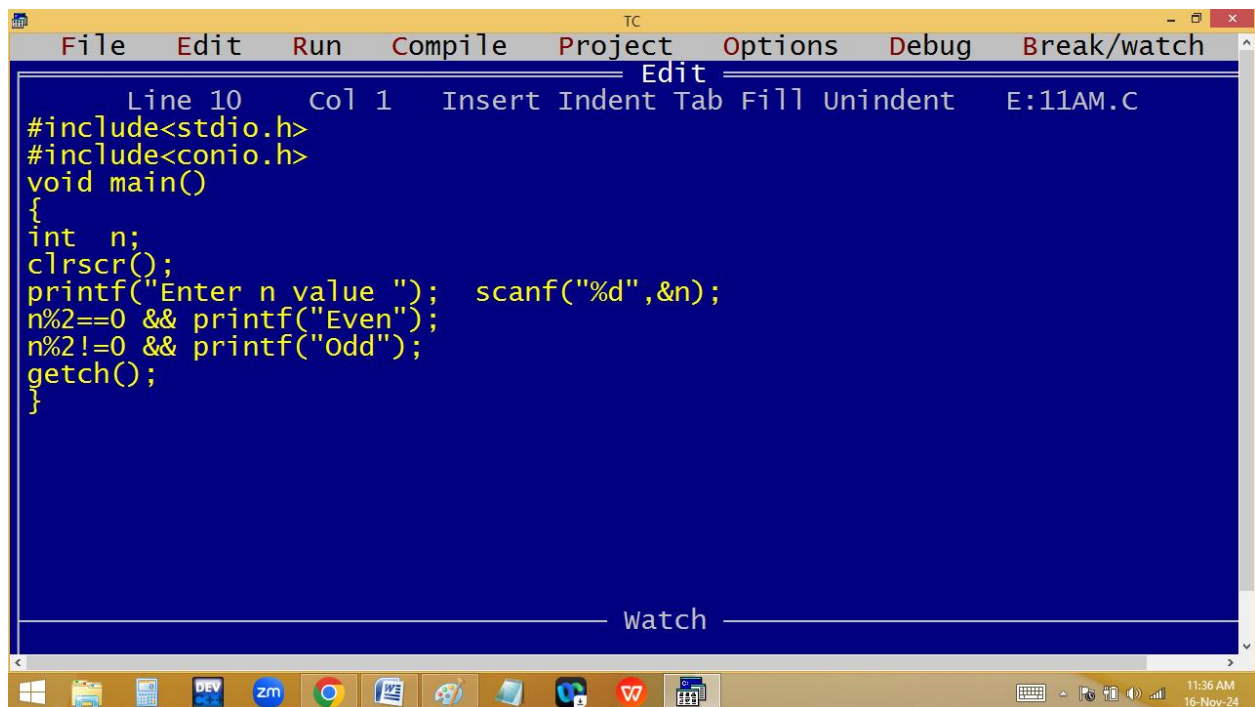
Diagram illustrating the logic of the ternary operator in the provided code snippet:

```
printf( n>0 ? "+Ve" : n<0 ? "-Ve" : "Zero" );
```

Arrows indicate the flow of execution for different values of `n`:

- `0 > 0` (False) leads to `0 < 0` (False), which leads to `"Zero"`.
- `4 > 0` (True) leads to `" +Ve"`.
- `-2 < 0` (True) leads to `" -Ve"`.
- `-2 > 0` (False) leads to `" -Ve"`.

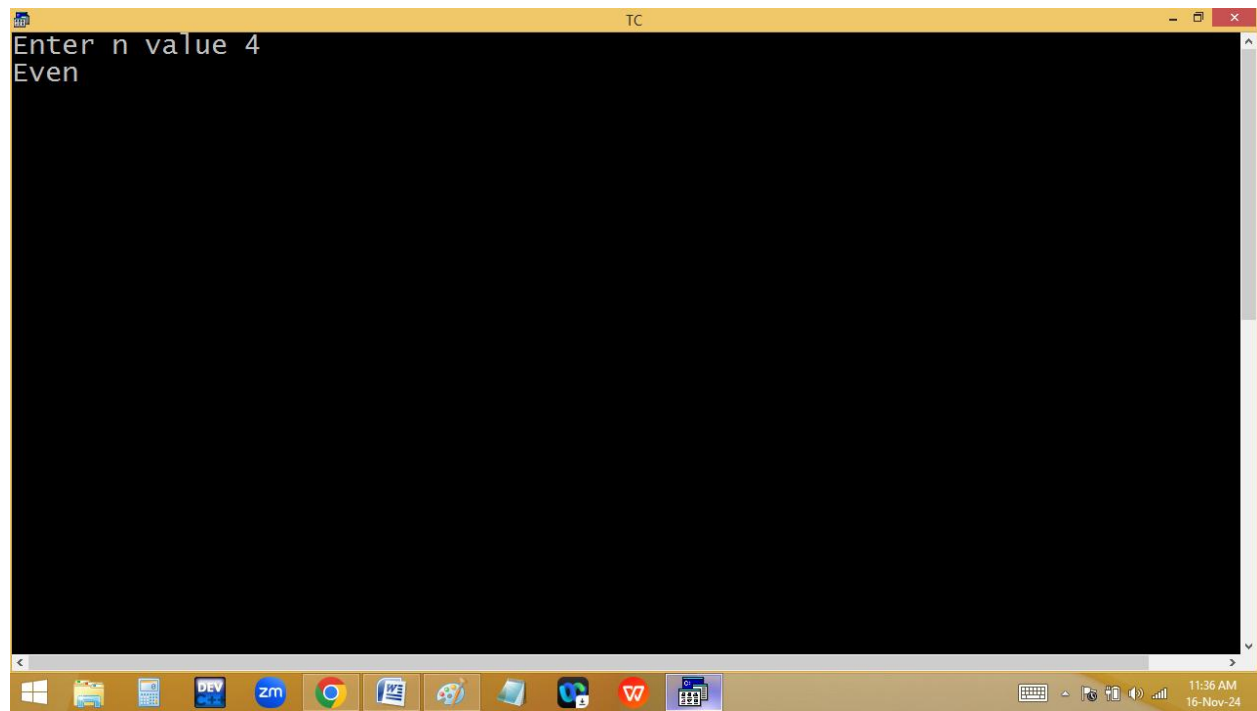
**Find even/odd without using if..else / ternary op:**



```
TC
File Edit Run Compile Project Options Debug Break/watch
Edit
Line 10 Col 1 Insert Indent Tab Fill Unindent E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
    int n;
    clrscr();
    printf("Enter n value "); scanf("%d",&n);
    n%2==0 && printf("Even");
    n%2!=0 && printf("Odd");
    getch();
}
```

Watch

11:36 AM 16-Nov-24



```
TC
Enter n value 5
Odd
```

$7\%2=1==0==>\text{false}$

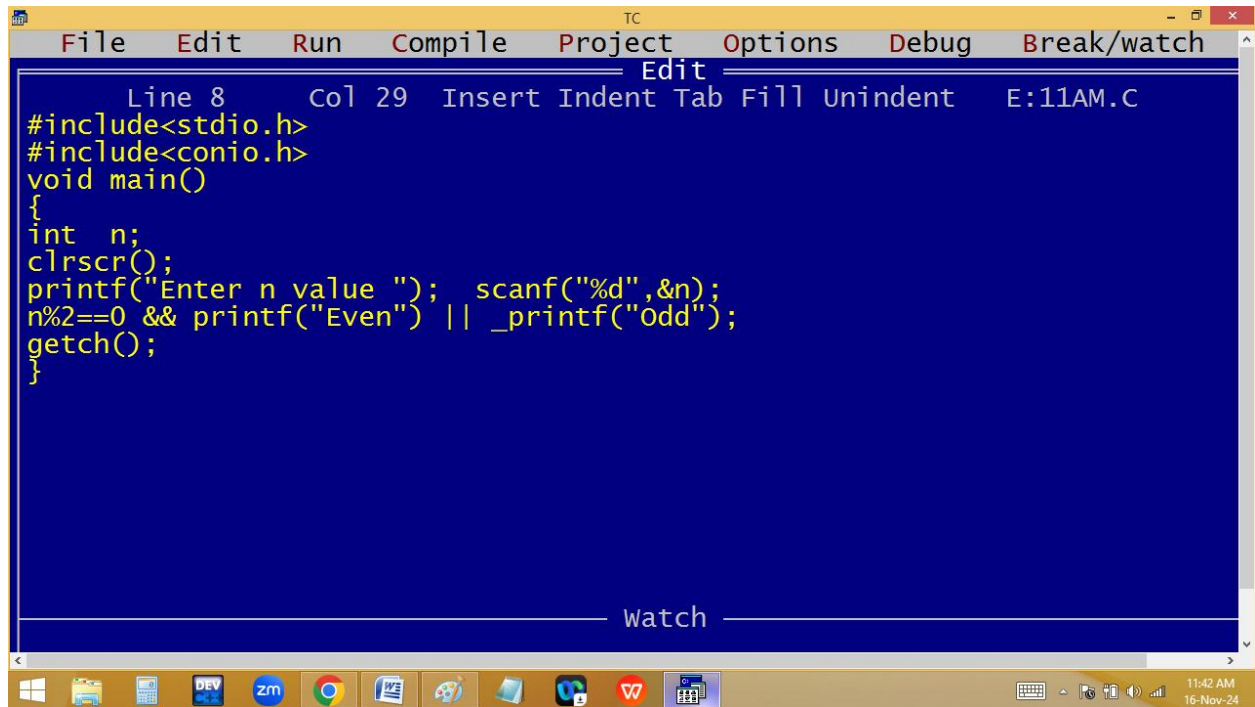
$4\%2=0==0$  

`n%2==0 && printf("even");`

$4\%2=0!=0==>\text{false}$  

`n%2!=0 && printf( "odd" );`

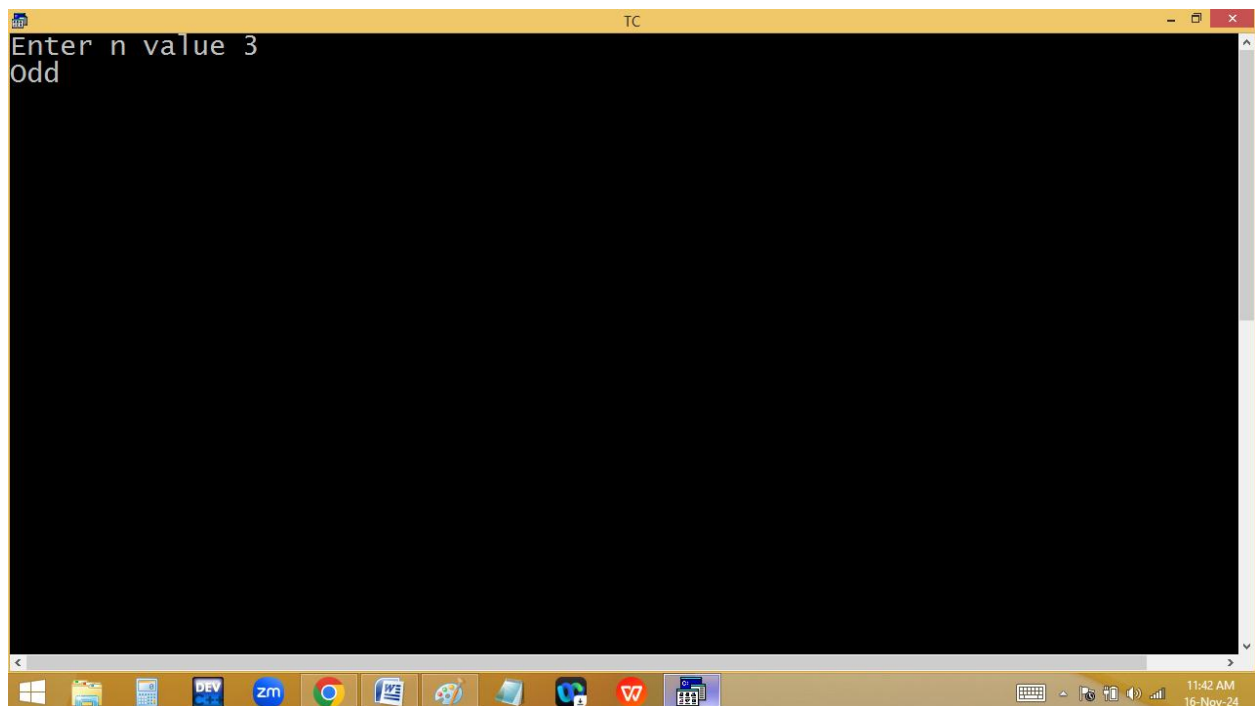
$7\%2=1!=0==>$  



The screenshot shows the Turbo C++ (TC) IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a toolbar. The main editing area has a blue background and contains the following C code:

```
Line 8 Col 29 Insert Indent Tab Fill Unindent E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
clrscr();
printf("Enter n value "); scanf("%d",&n);
n%2==0 && printf("Even") || _printf("Odd");
getch();
}
```

Below the code editor is a 'Watch' window. The Windows taskbar at the bottom shows various application icons and the system clock indicating 11:42 AM on 16-Nov-24.



The screenshot shows the Turbo C++ (TC) IDE with the same menu bar and toolbar. The main editing area now displays the program's output:

```
Enter n value 3
Odd
```

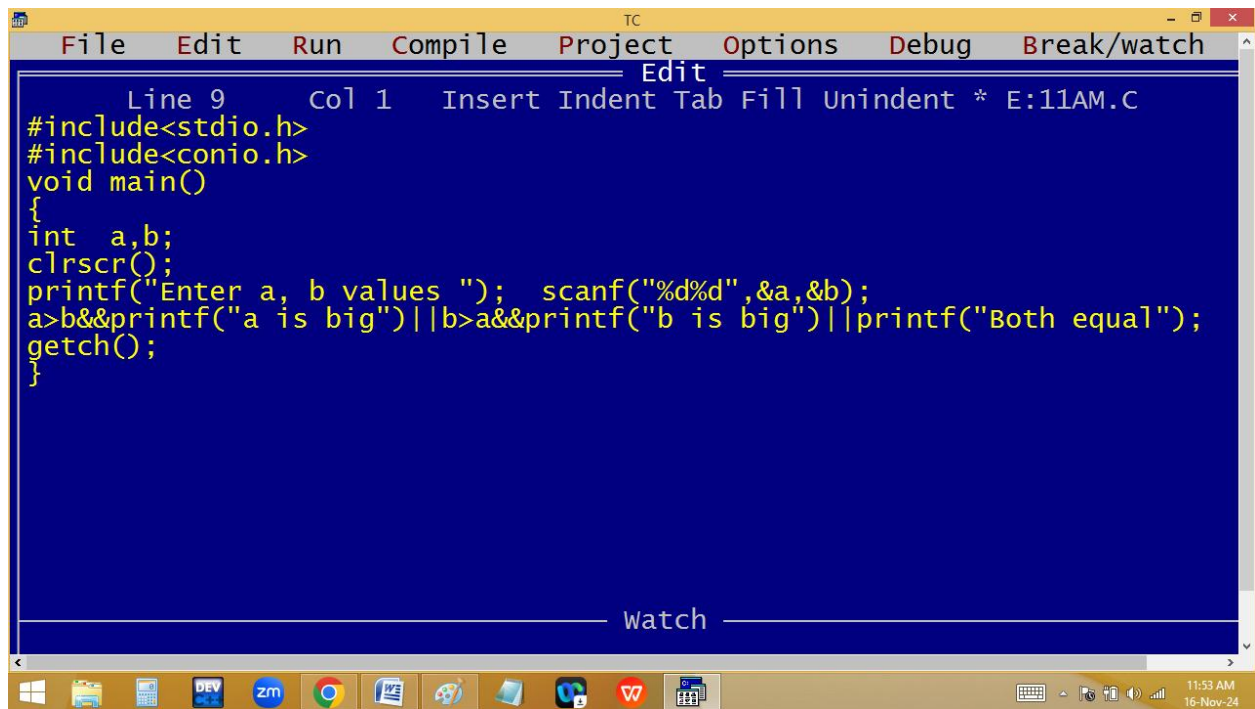
The rest of the IDE interface, including the Windows taskbar with the system clock at 11:42 AM on 16-Nov-24, remains the same.

```
TC
Enter n value 4
Even
```

$4\%2=0==0$  true X  
`n%2==0 && printf("even") || printf("odd");`

$5\%2=1==0 \Rightarrow \text{false}$  X ↗

**Finding max in 2 no's without using if..else/ternary op?**



The screenshot shows the Turbo C++ (TC) IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a toolbar. The main editing area has a blue background and contains the following C code:

```
Line 9 Col 1 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b;
clrscr();
printf("Enter a, b values "); scanf("%d%d",&a,&b);
a>b&&printf("a is big")||b>a&&printf("b is big")||printf("Both equal");
getch();
}
```

At the bottom of the window, there is a 'Watch' section and a Windows taskbar with various application icons and a system clock showing 11:53 AM on 16-Nov-24.



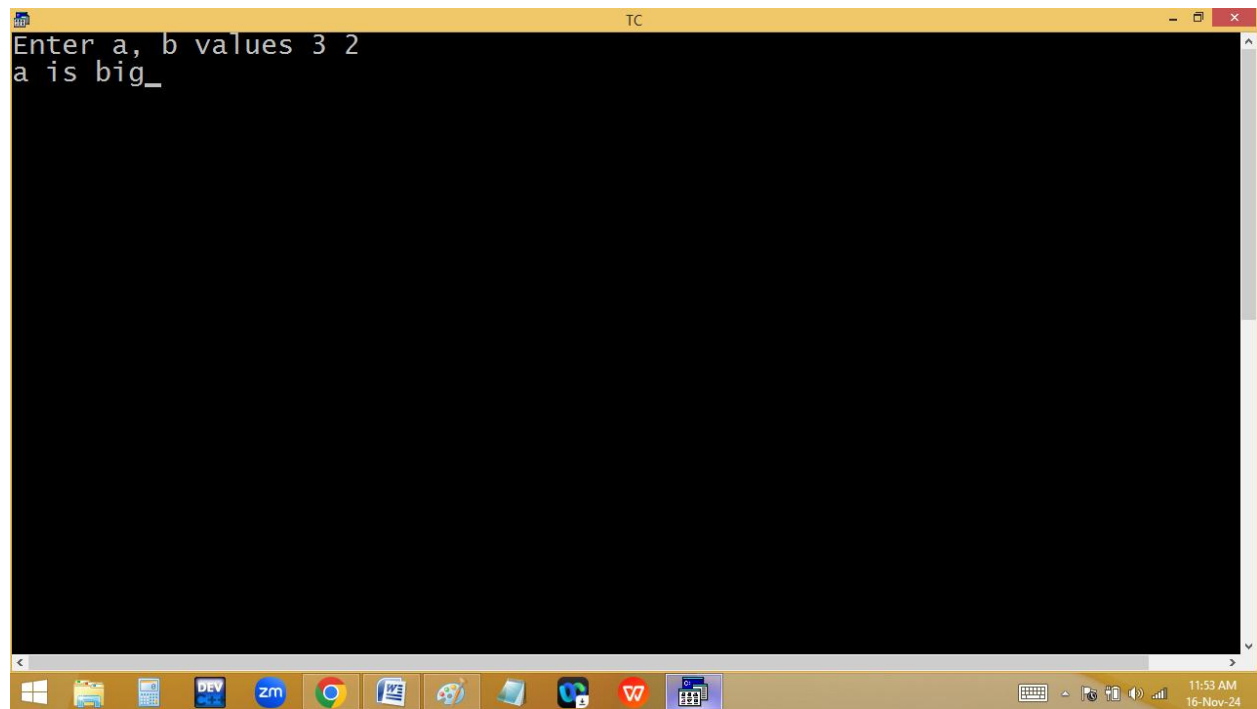
The screenshot shows the Turbo C++ (TC) IDE after the program has been executed. The main editing area now displays the output of the program:

```
Enter a, b values 1 2
b is big
```

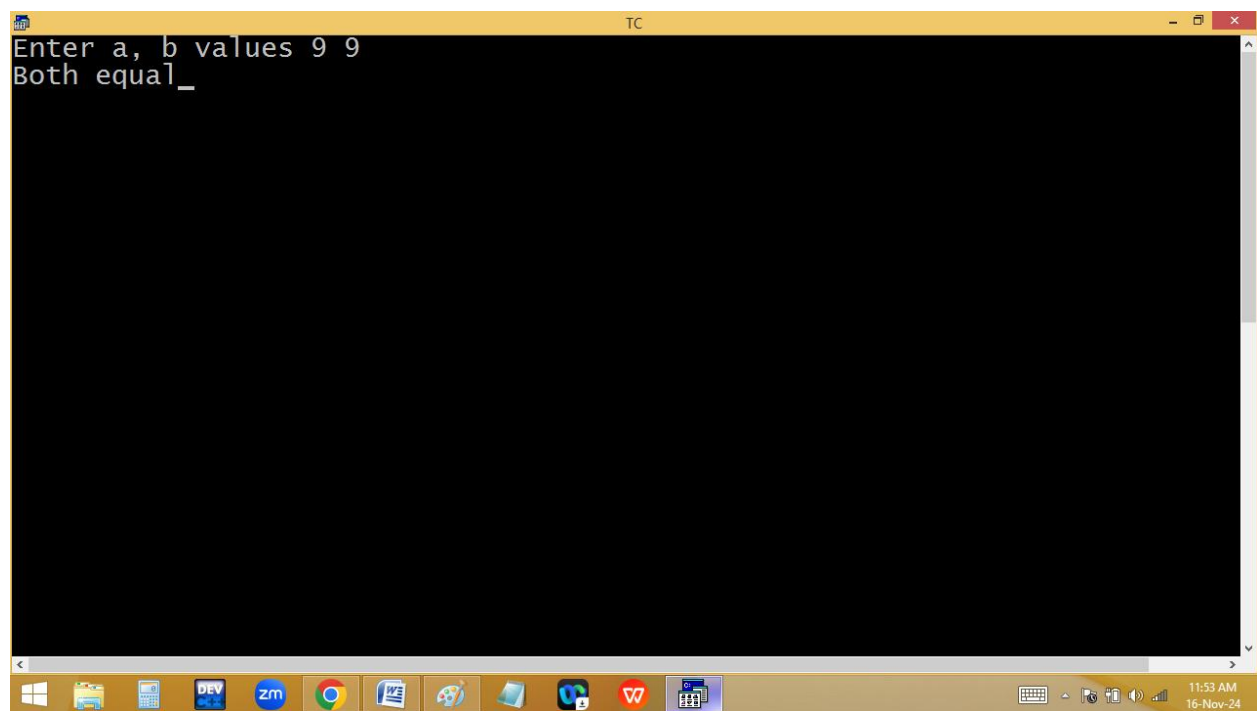
The rest of the IDE interface, including the menu bar, toolbar, and Windows taskbar, remains the same as in the first screenshot.



```
TC
Enter a, b values 3 2
a is big_
```



```
TC
Enter a, b values 9 9
Both equal_
```



9>9==>>false X      9>9==>>false X   
a>b&& p("a is big") || b>a&& p("b is big") || p("Both are equal");  
4>2==>>true 

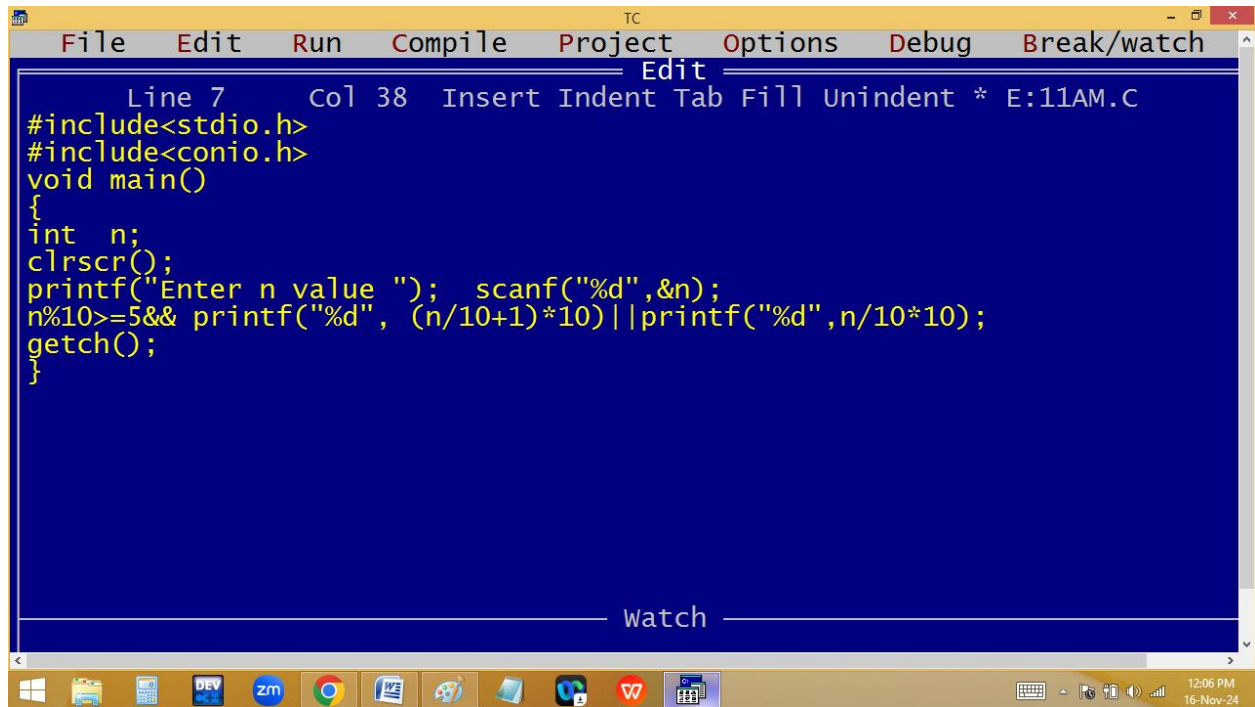
Read a no and if the last digit of given no is  $\geq 5$  go for next multiple otherwise go for previous multiple without using if..else / ternary op

257 last digit is 7 which is  $\geq 5 \rightarrow 260$

254 last digit is 4 which is  $\leq 5 \rightarrow 250$

  
257.51 ==> 258

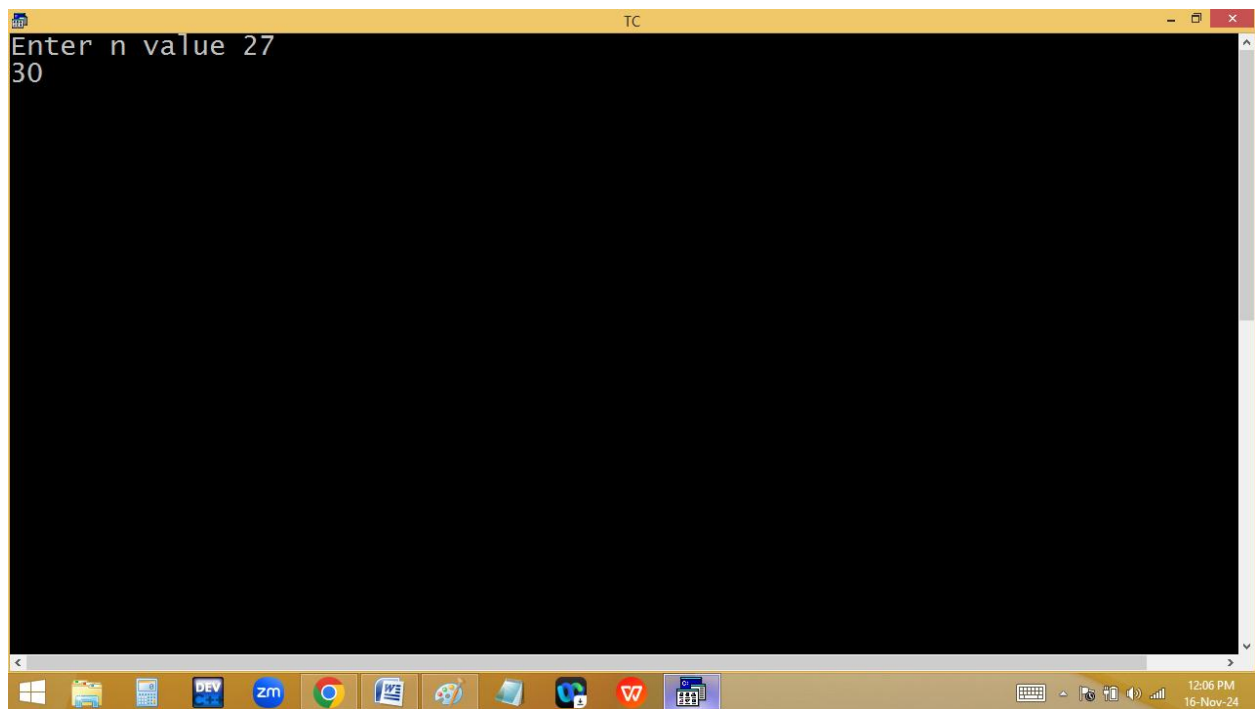
~~257.49 ==> 257~~



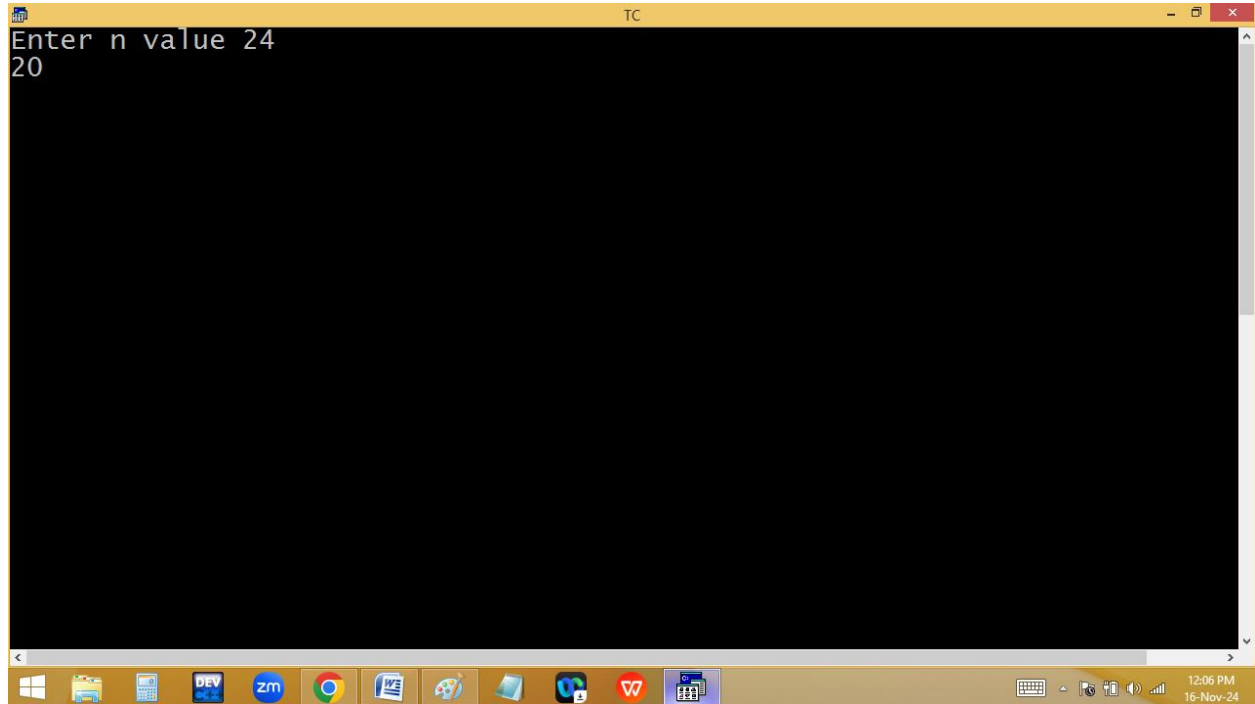
The screenshot shows the Turbo C++ (TC) IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a toolbar. The main editing area has a blue background and contains the following C code:

```
Line 7 Col 38 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
    int n;
    clrscr();
    printf("Enter n value "); scanf("%d",&n);
    n%10>=5&& printf("%d", (n/10+1)*10)||printf("%d",n/10*10);
    getch();
}
```

At the bottom of the window, there is a 'Watch' section which is currently empty. The Windows taskbar at the bottom shows various application icons and the system clock indicating 12:06 PM on 16-Nov-24.



This screenshot shows the same Turbo C++ IDE after the program has been executed. The output window at the top displays the text "Enter n value 27" followed by "30" on the next line. The rest of the IDE interface, including the menu bar and taskbar, remains the same as in the previous image.



```
Enter n value 24
20
```

## **Increment / Decrement / modify operators [ ++/-- ]**

They are used to increment / decrement a variable value by 1.

Eg:

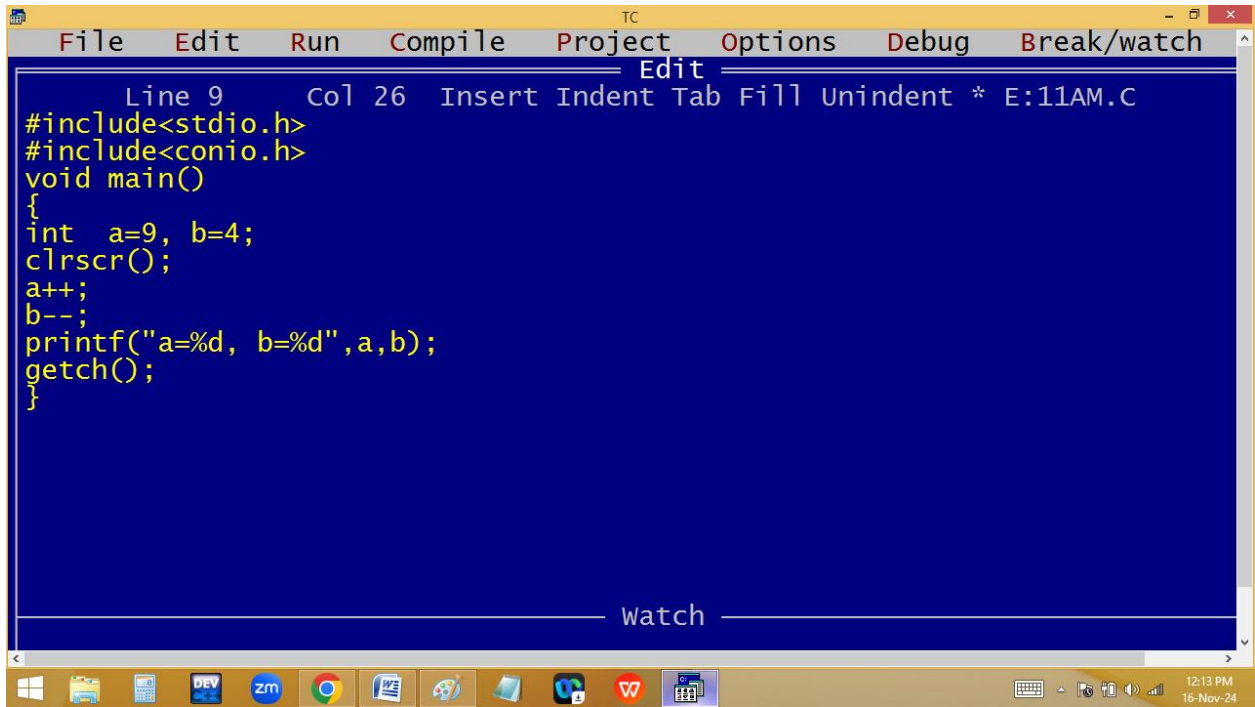
**Int a=9, b=4;**

**a++ → i.e. a=a+1 → a=10**

**b-- → i.e. b=b-1 → b=3**

**a+++; → Error**

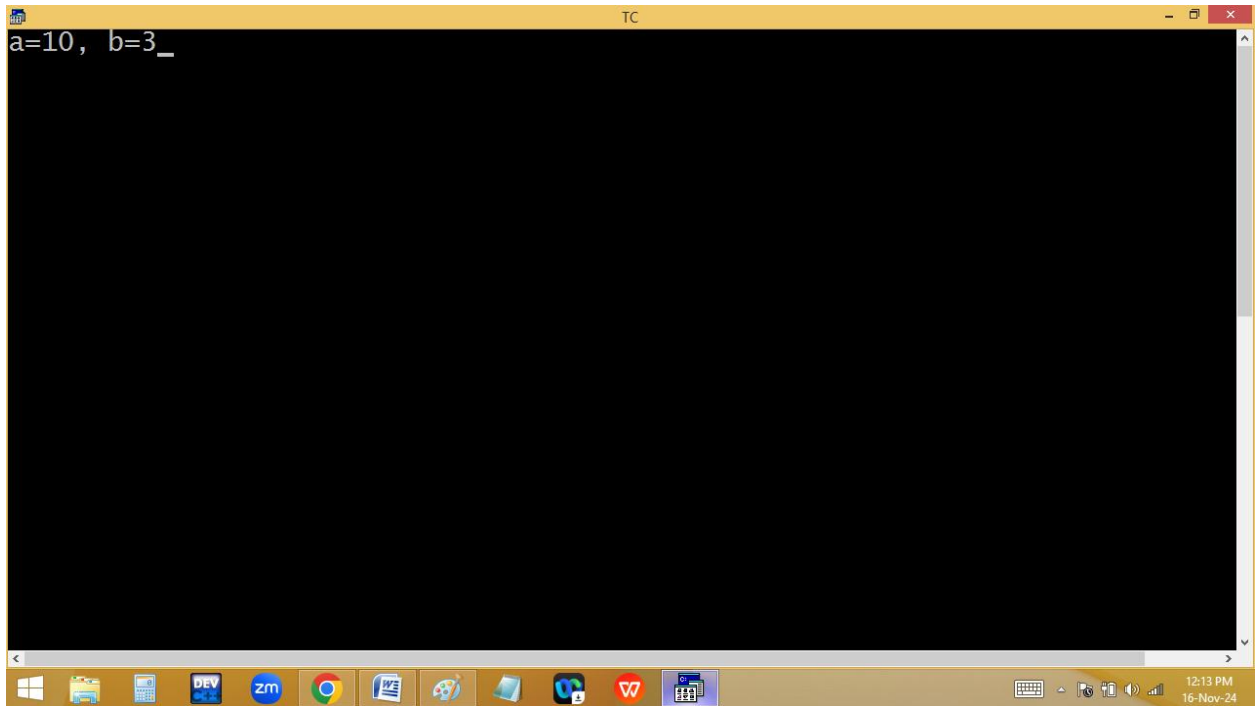
**b----; ➔ Error**



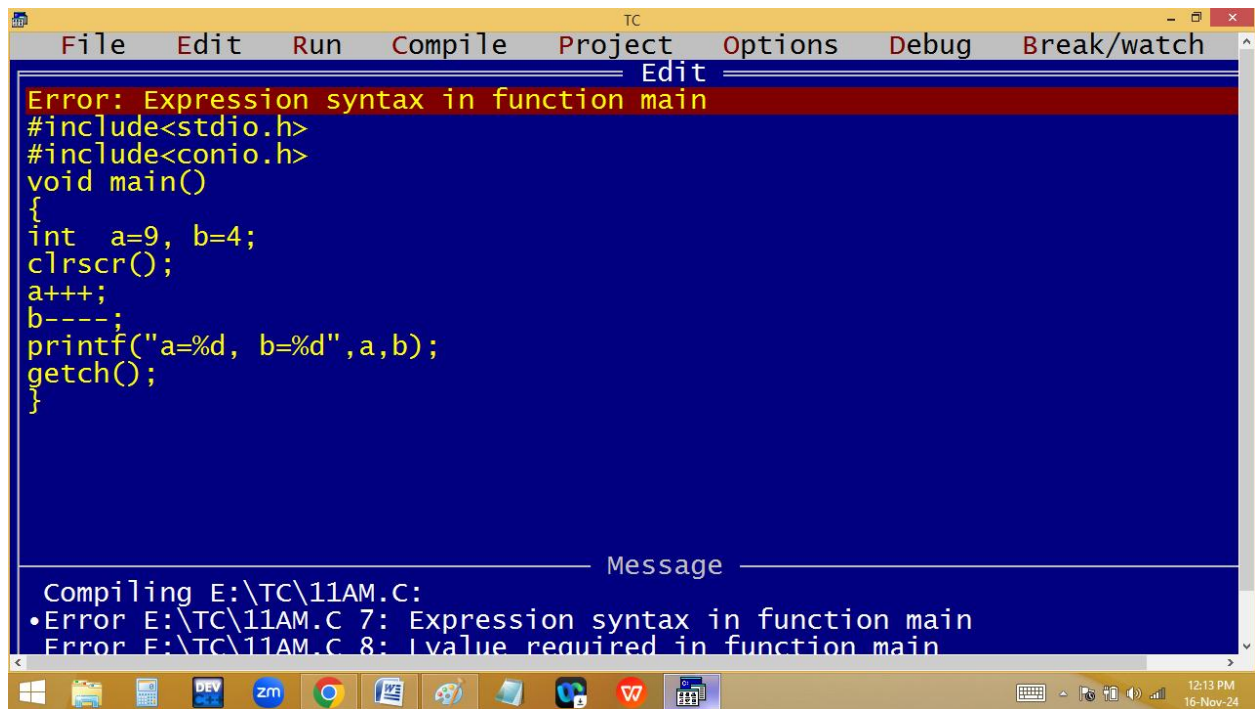
The screenshot shows the Turbo C++ (TC) editor window. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 9 Col 26 Insert Indent Tab Fill Unindent \* E:11AM.C'. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a=9, b=4;
    clrscr();
    a++;
    b--;
    printf("a=%d, b=%d",a,b);
    getch();
}
```

The cursor is positioned at the end of the line 'b--;'. Below the code area is a 'Watch' section. The Windows taskbar at the bottom shows the time as 12:13 PM on 16-Nov-24.



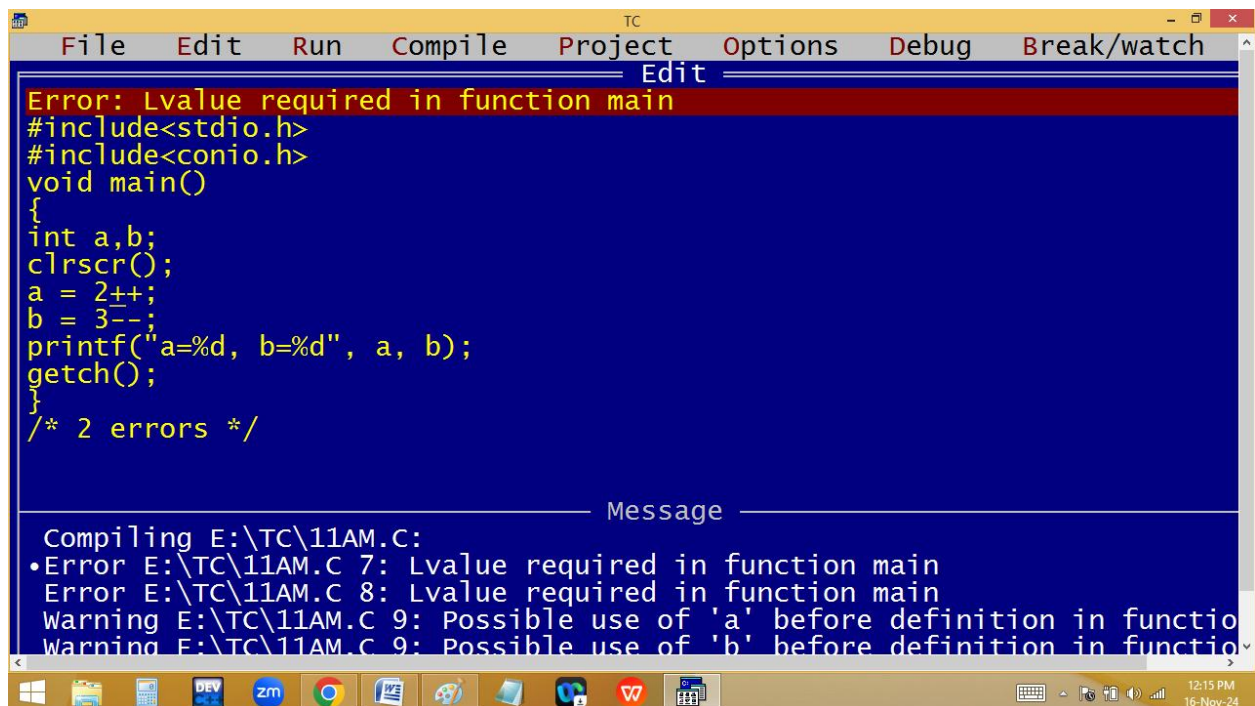
The screenshot shows the Turbo C++ (TC) console window. The output displayed is 'a=10, b=3\_'. The Windows taskbar at the bottom shows the time as 12:13 PM on 16-Nov-24.



The screenshot shows the Turbo C++ IDE with a C program in the editor. The program includes `<stdio.h>` and `<conio.h>`, and defines a `main` function. Inside the function, it declares `int a=9, b=4;`, calls `clrscr();`, increments `a` with `a++;`, decrements `b` with `b--;`, prints the values with `printf("a=%d, b=%d",a,b);`, and waits for input with `getch();`. The compiler has generated two error messages: "Error E:\TC\11AM.C 7: Expression syntax in function main" and "Error E:\TC\11AM.C 8: Lvalue required in function main".

```
File Edit Run Compile Project Options Debug Break/watch
Error: Expression syntax in function main
#include<stdio.h>
#include<conio.h>
void main()
{
int a=9, b=4;
clrscr();
a++;
b--;
printf("a=%d, b=%d",a,b);
getch();
}

Message
Compiling E:\TC\11AM.C:
•Error E:\TC\11AM.C 7: Expression syntax in function main
Error E:\TC\11AM.C 8: Lvalue required in function main
```



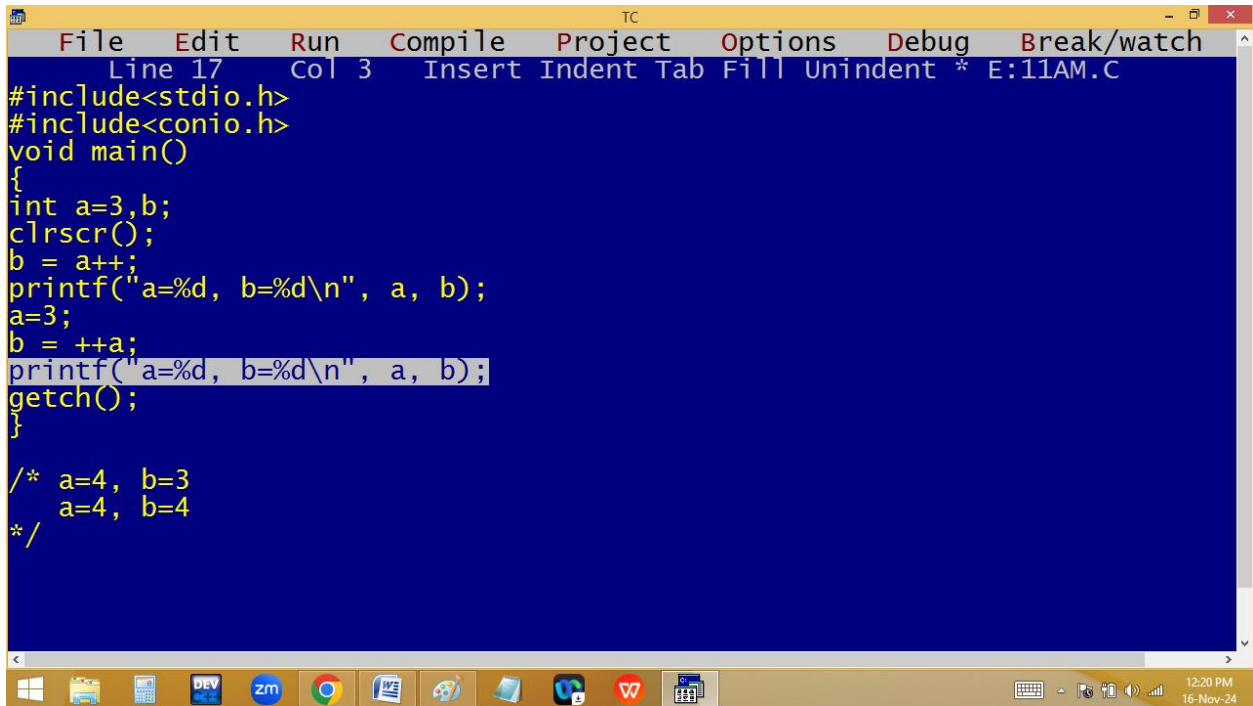
The screenshot shows the Turbo C++ IDE with a C program. The program declares `int a,b;` and calls `clrscr();`. It then attempts to increment `a` with `a = 2++;` and decrement `b` with `b = 3--;`, which causes "Lvalue required" errors. It then prints the values with `printf("a=%d, b=%d", a, b);` and waits for input with `getch();`. The compiler generates two "Error E:\TC\11AM.C 7: Lvalue required in function main" and "Error E:\TC\11AM.C 8: Lvalue required in function main" messages, along with two warnings about the possible use of `a` and `b` before their definitions. A comment `/* 2 errors */` is at the end of the program.

```
File Edit Run Compile Project Options Debug Break/watch
Error: Lvalue required in function main
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b;
clrscr();
a = 2++;
b = 3--;
printf("a=%d, b=%d", a, b);
getch();
}
/* 2 errors */

Message
Compiling E:\TC\11AM.C:
•Error E:\TC\11AM.C 7: Lvalue required in function main
Error E:\TC\11AM.C 8: Lvalue required in function main
Warning E:\TC\11AM.C 9: Possible use of 'a' before definition in function main
Warning E:\TC\11AM.C 9: Possible use of 'b' before definition in function main
```

**a++ meaning a = a + 1**

**2++ meaning  $2 = 2 + 1 \Rightarrow 2 = 3 \Rightarrow 2$  is a constant  $\Rightarrow$  error**



```
File Edit Run Compile Project Options Debug Break/watch
Line 17 Col 3 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=3,b;
clrscr();
b = a++;
printf("a=%d, b=%d\n", a, b);
a=3;
b = ++a;
printf("a=%d, b=%d\n", a, b);
getch();
}

/* a=4, b=3
   a=4, b=4
*/
```

### postfix increment:

a=3  
b=a++  
priority: =, a++  
1. b=a ==> **b=3**  
2. a++ ==> **a=4**

### prefix increment:

a=3  
b=++a;  
priority: ++a, =  
1. ++a==>**a=4**  
2. b=a==>**b=4**

**Note: prefix is always first and postfix is always last.**

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 15 Col 12 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=3,b;
clrscr();
b = a--;
printf("a=%d, b=%d\n", a, b);
a=3;
b = --a;
printf("a=%d, b=%d\n", a, b);
getch();
}

/* a=2, b=3
   a=2, b=2
*/
```

### postfix decrement:

a=3  
b=a--;  
priority: =, a--  
1. b=a==>**b=3**  
2. a-- ==>**a=2**

### prefix decrement:

a=3  
b=--a;  
priority: --a, =  
1. --a ==> **a=2**  
1. b=a==>**b=2**



```
File Edit Run Compile Project Options Debug Break/watch
Line 17 Col 3 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=3,b;
clrscr();
a = a--;
printf("a=%d\n", a);
a=3;
a = --a;
printf("a=%d\n", a);
getch();
}

/* a=2
   a=2
*/
```

a=3  
a=a--;  
priority: =, a--  
1. a=a ==> a=3  
2. a-- ==> **a=2**

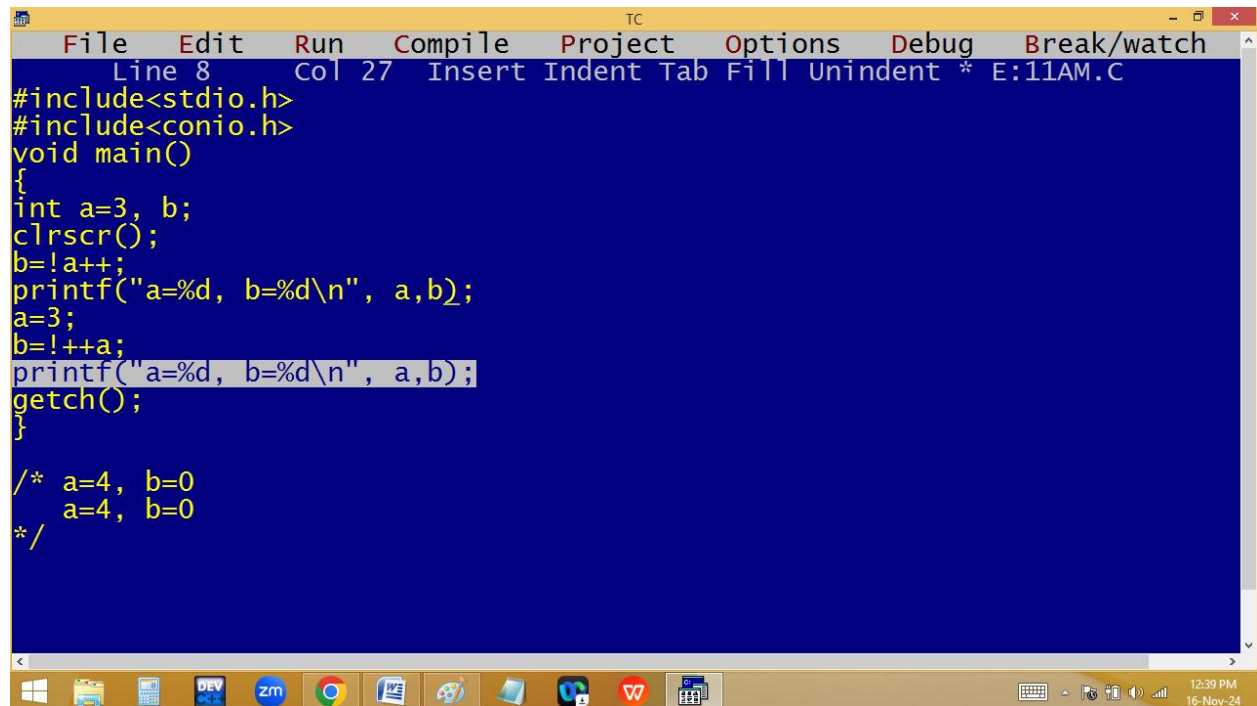
a=3  
a=--a;  
priority: --a, =  
1. --a ==> a=2  
2. a=a ==> **a=2**

**Note: Until assigning to any other variable pre and post are same.**

```
File Edit Run Compile Project Options Debug Break/watch
Line 16 Col 7 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=3;
clrscr();
a++;
printf("a=%d\n", a);
a=3;
++a;
printf("a=%d\n", a);
getch();
}

/* a=4
   a=4_
*/
```

```
a=4
a=4
_
```



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 8 Col 27 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=3, b;
clrscr();
b=!a++;
printf("a=%d, b=%d\n", a,b);
a=3;
b=!++a;
printf("a=%d, b=%d\n", a,b);
getch();
}

/* a=4, b=0
a=4, b=0
*/
```

12:39 PM  
16-Nov-24

